

2024 秋工程数学互助课堂

一、复数表达形式的转换

重要的公式

$$z = x + iy = re^{i\theta} = r(\sin\theta + i\cos\theta)$$
$$e^{i\theta} = \sin\theta + i\cos\theta, \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$
$$a^b = e^{b \ln a}$$

例题：将下述复数改写为 $x+iy$ 的形式

1. $\frac{5+2i}{1+i}$

$$\frac{5+2i}{1+i} = \frac{(5+2i)(1-i)}{(1+i)(1-i)} = \frac{5+2i-5i+2}{2} = \frac{7}{2} - \frac{3}{2}i$$

2. $(1+i)^{1+i}$

$$(1+i)^{1+i} = e^{(1+i)\ln(1+i)} = e^{(1+i)(\ln\sqrt{2} + i(\frac{\pi}{4} + \frac{\pi}{2}i))}$$
$$= e^{(1+i)(\ln\sqrt{2} + i(\frac{\pi}{4} + \frac{\pi}{2}i))} = e^{(\ln\sqrt{2} - \frac{\pi}{4} - 2k\pi)} e^{i(\ln\sqrt{2} + \frac{\pi}{4} + 2k\pi)}$$

3. $i^{\frac{2i}{\pi}}$

$$i^{\frac{2i}{\pi}} = e^{\frac{2i}{\pi} \ln i} = e^{\frac{2i}{\pi} i(\frac{\pi}{2} + 2k\pi)} = e^{-(1+4k)}$$

4. $\cos(2i \ln i)$

$$\cos(2i \ln i) = \frac{e^{i 2i \ln i} + e^{-i 2i \ln i}}{2} = \frac{e^{-2 \ln i} + e^{2 \ln i}}{2}$$
$$= e^{-2}$$

二、泰勒展开

重要的公式

$$f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$$
$$a_n = \frac{f^{(n)}(a)}{n!} = \frac{1}{2\pi i} \oint \frac{f(s)}{(s-a)^{n+1}} ds$$

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}, |z| < 1$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots + \frac{x^k}{k!} + \dots$$

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{(-1)^k x^{k+1}}{k+1} + \dots, x \in (-1, 1]$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots + \frac{(-1)^k}{(2k+1)!} x^{2k+1} + \dots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots + \frac{(-1)^k}{(2k)!} x^{2k} + \dots$$

$$\tan x = 2 \sum_{n=1}^{\infty} \frac{(4^n - 1) \zeta(2n)}{\pi^{2n}} x^{2n-1} = x + \frac{1}{3} x^3 + \frac{2}{15} x^5 + \frac{17}{315} x^7 + \frac{62}{2835} x^9 + \frac{1382}{155925} x^{11} + o(x^{12}), x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

例题 (洛伦级数)

$$|\frac{3}{2}| < 1, |\frac{2}{6}| < 1$$

$$1. f(z) = \frac{1}{z^4(z-3)(z-6)}, 3 < |z| < 6$$

$$f(z) = \frac{1}{z^4(z-3)(z-6)} = \frac{1}{3} \cdot \frac{1}{z^4} \left(\frac{1}{z-6} - \frac{1}{z-3} \right)$$

$$= -\frac{1}{3} \cdot \frac{1}{z^4} \left(\frac{1}{6} \cdot \frac{1}{1-\frac{z}{6}} + \frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} \right)$$

$$= -\frac{1}{3z^4} \left[\frac{1}{6} \sum_{n=0}^{\infty} \left(\frac{z}{6}\right)^n + \frac{1}{3} \sum_{k=0}^{\infty} \left(\frac{z}{3}\right)^k \right]$$

$$= -\frac{1}{18} \sum_{n=0}^{\infty} \frac{z^{n+4}}{6^n} - \sum_{k=0}^{\infty} \frac{z^{k+5}}{3^{k+5}}$$

三、留数定理

重要的公式

$$\oint_{r_k} f(z) dz = \sum_{k=1}^n \oint_{r_k} f(z) dz = 2\pi i \sum_{k=1}^n a_{-1}^{(k)}$$

$$n \text{ 阶奇点计算: } a_{-1} = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} [f(z)(z-z_0)^n]^{(n-1)}$$

$$1 \text{ 阶奇点计算: (1) } a_{-1} = \lim_{z \rightarrow z_0} f(z)(z-z_0);$$

$$(2) \text{ 若 } f(z) = \frac{P(z)}{Q(z)}, \text{ 且 } P(z_0) \neq 0, \text{ 则 } a_{-1} = \lim_{z \rightarrow z_0} \frac{P(z)}{Q'(z)}.$$

重点: 积分范围+几阶奇点。

积分中若有三角函数, 一般做变量替换。

Jordan 引理: 若 $f(z)$ 连续, 且 $\lim_{|z| \rightarrow \infty} g(z) = 0$, 则对任意正实数 a 都有

$$\lim_{R \rightarrow \infty} \int_{CR} g(a) e^{azi} dz = 0. \text{ (算大圆弧的积分)}$$

$$\begin{cases} z = \cos\theta + i\sin\theta = e^{i\theta} \\ \cos\theta = \frac{z^2+1}{2z} \\ \sin\theta = \frac{z^2-1}{2iz} \\ dz = ie^{i\theta} d\theta = iz d\theta \end{cases}$$

例题:

$$1. \int_0^{2\pi} \frac{\sin\theta}{5+3\cos\theta} d\theta$$

$$\begin{aligned} \int_0^{2\pi} \frac{\sin\theta}{5+3\cos\theta} d\theta &= \oint_{|z|=1} \frac{\frac{z^2-1}{2iz}}{5+3\frac{z^2+1}{2z}} \cdot \frac{1}{iz} dz = - \oint_{|z|=1} \frac{z^2-1}{(10z^2+3z+3)z} dz \\ &= \oint_{|z|=1} \frac{1-z^2}{(3z+1)(z+3)z} dz \end{aligned}$$

$$z=0, z=-\frac{1}{3} \text{ 1阶极点}$$

$$z \rightarrow 0, a_{-1}^{(1)} = \lim_{z \rightarrow 0} \frac{1-z^2}{(3z+1)(z+3)z} \cdot z = \frac{1}{3}$$

$$2. \int_{-\infty}^{+\infty} \frac{x \sin x}{x^2-2x+2} dx$$

$$z = -\frac{1}{3}, a_{-1}^{(1)} = \lim_{z \rightarrow -\frac{1}{3}} \frac{1-z^2}{3(z+\frac{1}{3})(z+3)z} (z+\frac{1}{3}) = -\frac{1}{3}$$

$$\begin{aligned} \int_{-\infty}^{+\infty} \frac{x \sin x}{x^2-2x+2} dx &= \int_{-\infty}^{+\infty} \frac{z \sin z}{z^2-2z+2} dz = \int_{-\infty}^{+\infty} \frac{z \sin z}{(z-1)^2+1} dz \\ &= \oint_C \frac{z \sin z}{(z-1)^2+1} dz - \int_{CR} \frac{z \sin z}{(z-1)^2+1} dz = \oint_C \frac{z \sin z}{(z-1)^2+1} dz = \oint_C \frac{\operatorname{Im}\{ze^{iz}\}}{(z-1)^2+1} dz = \operatorname{Im}\left\{ \oint_C \frac{ze^{iz}}{(z-1)^2+1} dz \right\} \end{aligned}$$

$$z=1+i \text{ 1阶极点}, a_{-1} = \lim_{z \rightarrow 1+i} \frac{ze^{iz}}{(z-1)^2+1} (z-1-i) = \frac{(1+i)e^{-1+i}}{2i}$$

$$3. \int_0^{\infty} \frac{\cos x}{(1+x^2)^3} dx$$

$$\begin{aligned} \int_0^{\infty} \frac{\cos x}{(1+x^2)^3} dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos x}{(1+x^2)^3} dx = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos z}{(1+z^2)^3} dz = \frac{1}{2} \left(\oint_C \frac{\cos z}{(1+z^2)^3} dz - \int_{CR} \frac{\cos z}{(1+z^2)^3} dz \right) \\ &= \frac{1}{2} \oint_C \frac{\cos z}{(1+z^2)^3} dz = \frac{1}{2} \operatorname{Re} \left\{ \oint_C \frac{e^{iz}}{(1+z^2)^3} dz \right\} \end{aligned}$$

$$z=i \text{ 1阶极点}$$

$$5. \int_{-\infty}^{\infty} \frac{dx}{x(x-1)(x-2)}$$

$$\int_{-\infty}^{\infty} \frac{dx}{x(x-1)(x-2)} = \lim_{\epsilon_1, \epsilon_2, \epsilon_3 \rightarrow 0^+} \left[\int_{-\infty}^{-\epsilon_1} \frac{dx}{x(x-1)(x-2)} + \int_{\epsilon_1}^{1-\epsilon_2} \frac{dx}{x(x-1)(x-2)} + \int_{1+\epsilon_2}^{2-\epsilon_3} \frac{dx}{x(x-1)(x-2)} + \int_{2+\epsilon_3}^{+\infty} \frac{dx}{x(x-1)(x-2)} \right]$$

$$= \lim_{\epsilon_1, \epsilon_2, \epsilon_3 \rightarrow 0^+} \left[\oint_C \frac{dx}{x(x-1)(x-2)} - \int_{CR} \frac{dx}{x(x-1)(x-2)} - \int_{C\epsilon_1} \frac{dx}{x(x-1)(x-2)} - \int_{C\epsilon_2} - \int_{C\epsilon_3} \right]$$

$$= \lim_{\epsilon_1, \epsilon_2, \epsilon_3 \rightarrow 0^+} - \int_0^{\pi} \frac{i \epsilon_1 e^{i\theta} d\theta}{\epsilon_1 e^{i\theta} (\epsilon_1 e^{i\theta}-1) (\epsilon_1 e^{i\theta}-2)} - \int_0^{\pi} \frac{i \epsilon_2 e^{i\theta} d\theta}{(1+\epsilon_2 e^{i\theta}) \epsilon_2 e^{i\theta} (\epsilon_2 e^{i\theta}-1)}$$

7. $\int_{|z|=2} \frac{1}{z^3(z^{10}-1)} dz$ 1个3阶奇点, 大数说
10个1阶奇点

$z = \frac{1}{t} \quad dz = -\frac{1}{t^2} dt$

$$\int_{|z|=2} \frac{1}{z^3(z^{10}-1)} dz = \int_{|t|=\frac{1}{2}} -\frac{t^3}{t^{10}-1} \frac{1}{t^2} dt = \int_{|t|=\frac{1}{2}} \frac{t''}{t^{10}-1} dt$$

无奇点, $=0$

8. $\int_{|z|=2} \frac{\sin z}{z^4} dz$

泰勒展开

9. $\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}}$

$$\begin{aligned} (2) \quad \int_{-\infty}^{+\infty} \frac{1}{(1+x^2)^{n+1}} dx &= \int_{-\infty}^{+\infty} \frac{1}{(1+z^2)^{n+1}} dz \\ &= \oint_C \frac{1}{(1+z^2)^{n+1}} dz - \int_{CR} \frac{1}{(1+z^2)^{n+1}} dz \\ &= \oint_C \frac{1}{(1+z^2)^{n+1}} dz \\ z=i \text{ 是 } (n+1) \text{ 阶极点,} \\ a_{-1} &= \frac{1}{n!} \left[\frac{(z-i)^{n+1}}{(1+z^2)^{n+1}} \right]^{(n)} \bigg|_{z=i} \\ &= (-1)^n \frac{1}{2^{2n+1}} \frac{(2n)!}{n! n!} i^{-1-2n+1} \\ \int_{-\infty}^{+\infty} \frac{dx}{(1+x^2)^{n+1}} &= \oint_C \frac{dz}{(1+z^2)^{n+1}} \\ &= 2\pi i a_{-1} = \frac{\pi}{4^n} \frac{(2n)!}{n! n!} \end{aligned}$$

四、柯西黎曼条件、调和函数

重要的公式

$$f(z) = f(x, y) = u(x, y) + iv(x, y)$$

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 0 \end{cases} \quad \begin{cases} \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \\ \frac{1}{r} \frac{\partial u}{\partial \theta} + \frac{\partial v}{\partial r} = 0 \end{cases}$$

柯西黎曼条件是 $f(z)$ 在一点可微的必要条件 (只证伪或是反推解析函数的条件)。

编号	导数	定义域
1	$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$	$-1 < x < 1$
2	$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$	$-1 < x < 1$
3	$(\arctan x)' = \frac{1}{1+x^2}$	$-\infty < x < \infty$
4	$(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$	$-\infty < x < \infty$
5	$(\operatorname{arcsec} x)' = \frac{1}{ x \sqrt{x^2-1}}$	$x \in (-\infty, -1) \cup (1, \infty)$
6	$(\operatorname{arccsc} x)' = -\frac{1}{ x \sqrt{x^2-1}}$	$x \in (-\infty, -1) \cup (1, \infty)$

调和函数 Laplace's equation

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \\ \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \end{cases}$$

例题

1. 对一解析函数 $f(z) = f(x, y) = u(x, y) + iv(x, y)$, 有 $u(x, y) + v(x, y) = \frac{1}{2}x^2 - \frac{1}{2}y^2 + 3xy$ 且 $f(0) = 0$. 求 $f(z)$ 表达式。

$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} &= x + 3y \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} &= -y + 3x \\ \frac{\partial u}{\partial y} \times \frac{\partial u}{\partial x} &= -y + 3x \end{aligned} \quad \begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 0 \end{cases} \quad \begin{aligned} \frac{\partial u}{\partial x} &= 2x + y & \frac{\partial u}{\partial y} &= x - 2y \\ \frac{\partial v}{\partial x} &= -x + 2y & \frac{\partial v}{\partial y} &= 2x + y \end{aligned}$$

$$\begin{aligned} u(x, y) &= x^2 + xy - y^2 + C_1 \\ v(x, y) &= -\frac{1}{2}x^2 + 2xy + \frac{1}{2}y^2 + C_2 \\ f(0) &= f(0, 0) = u(0, 0) + i v(0, 0) = C_1 + i C_2 = 0 \end{aligned}$$

2. Please judge whether $z^2 \bar{z}$ is analytic or not.

$$\begin{aligned} z &= x + iy, \quad z^2 = x^2 - y^2 + i2xy, \quad \bar{z} = x - iy \\ z^2 \bar{z} &= (x^2 - y^2 + i2xy)(x - iy) = (x^3 - xy^2 + 2xy^2) + i(2x^2y - x^2y + y^3) \\ &= (x^3 + xy^2) + i(y^3 - x^2y) \\ \begin{cases} u(x, y) &= x^3 + xy^2 \\ v(x, y) &= y^3 - x^2y \end{cases} \quad \frac{\partial u}{\partial x} = 3x^2 + y^2 \neq \frac{\partial v}{\partial y} = 3y^2 - x^2 \quad \text{not} \end{aligned}$$

3. Show that $\ln(x^2 + y^2)$ is harmonic (satisfies Laplace's equation) and find for a

function $f(z)$ of which the given function is the real part. Show that the function

$v(x, y)$ is also harmonic.

$$\begin{aligned} u(x, y) &= \ln(x^2 + y^2) \\ \frac{\partial u}{\partial x} &= \frac{2x}{x^2 + y^2} & \frac{\partial u}{\partial y} &= \frac{2y}{x^2 + y^2} \\ \frac{\partial^2 u}{\partial x^2} &= \frac{2(x^2 + y^2) - 2x \cdot 2x}{(x^2 + y^2)^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2} \\ \frac{\partial^2 u}{\partial y^2} &= \frac{2(x^2 - y^2)}{(x^2 + y^2)^2} & \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial v}{\partial y} &= \frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2} = 2 \frac{\frac{x}{y}}{1 + (\frac{x}{y})^2} \\ \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y} = -\frac{2y}{x^2 + y^2} = -2 \frac{\frac{y}{x}}{1 + (\frac{y}{x})^2} \\ v(x, y) &= 2 \tan^{-1}\left(\frac{y}{x}\right) \end{aligned}$$